

Coarse-Graining Dynamics and the Emergence of Complexity in Evolved Intelligence

Krohn–Rhodes Complexity & Evolution

Main Idea: *Krohn–Rhodes theory reveals hierarchical computational structure in automata and provides a framework for testing whether evolution drives systems toward maximal complexity.*

Key Points:

- Finite automata can be decomposed into hierarchical layers of:
 - **Permutation groups** → reversible behavior
 - **Reset mappings** → irreversible behavior
- This decomposition gives a **coarse-grained measure of structural complexity**.
- **Essential dependencies** arise when certain state subsets require higher-level group actions—forcing specific groups to appear at distinct levels.
- **Rhodes' Conjecture on the Evolution-Complexity Relations:** Evolved automata (or organisms) with N states tend to approach **maximal complexity** $\approx N - 1$.
- Rationale: if many states are maintained, evolution may “make them work harder” by increasing structural complexity.
- Competing hypothesis: **evolution may instead favor minimal complexity** if simpler solutions suffice or if complexity carries costs.
- This work **empirically tests** how evolved automata balance memory (state count) and complexity across different tasks and environments.
- Evolutionary methods follow the general classical framework of evolutionary algorithms of *Fogel et al. (1966) evolving Mealy automata* and variants (evolution strategies) applying Darwinian evolution to populations of automata agents solving computational intelligence tasks.

Krohn Rhodes theory => decomposition theory

Configuration space

Identity of a group implies a semigroup

An irreversible group is a semigroup

Coordinate system via Krohn Rhodes and navigating on that system.

Provides the computational structure of an automata, giving the coordinate system, course to finegraining,

Semigroup acting on state space.

Frobenius lagrange nehaniv 2009

Free semigroup.

Mod counter in coordinate system like lac operaon

Compexlity measure in korhn rodhes

Mealyu automata

Fogel 1966

Finite simple gorpus are connected with error correction in the presence of noise.

Semigroups, Monoids, group:

A set must be defined as a set of elements with the property X, s.t. **closure** can be satisfied within the set.

- **Semigroup (associativity):** A nonempty set S with a mapping μ from the cartesian product $S \times S$ into S forms a semigroup if **the binary operation μ satisfies the associative law: $\mu(s1, \mu(s2, s3)) = \mu(\mu(s1, s2), s3)$ for all $s1, s2, s3 \in S$. The mapping μ is called multiplication in S , and $\mu(s1, s2)$ is the product of $s1$ and $s2$ (in that order).**
- One writes instead of $s1 = \mu(s2, s3)$ simply $s1 = s2s3$ ($s1, s2, s3 \in S$). Associativity guarantees that products written without parentheses have well-defined values in S .
- For any pair $H1, H2 \subseteq S$, we write $H1H2 = \{h1h2 \mid h1 \in H1, h2 \in H2\}$
- **Subsemigroup (of group S):** if $S' \subseteq S$, and for every $s1, s2 \in S'$, $s1s2 \in S'$.
 - o The subsemigroup of S generated by a subset $H \subseteq S$ is $\langle H \rangle = \bigcup_{n \geq 1} H^n$
 - This definition seems to be such that two arbitrary subset $H1, H2 \in S$ (semigroup) does not form a subsemigroup as it doesn't satisfy $h1h2 \in S$, thus $H1 = H2$.
 - o A subsemigroup has closure amongst associativity in the subsemigroup.
 - o Associativity of S guarantees that $H1(H2H3) = (H1H2)H3$ for every choice of subsets $H1, H2, H3 \subseteq S$
- Right-zero element: an element $r \in S$ if $sr = r$ for all $s \in S$.
- Left-zero element, $l \in S$, if $ls = l$ for all $s \in S$.
- Zero element, $o \in S$, if $os = so = o$, $s \in S$.
- If the semigroup has both a left-zero element and a right-zero element r , then it has an unambiguously determined zero element $o = lr = l = r$.
- **Monoid (identity):** A semigroup s.t. an identity element 1 exists for which $s1 = 1s = s$, $s \in S$.
 - o Both the zero and identity elements are uniquely determined and clearly idempotents.

- **Group (invertibility):** A group where if for every $s \in S$ there is an $s^{-1} \in S$ such that $ss^{-1} = s^{-1}s = 1$.
 - o Inverses of a group are unique: since if $t, r \in S$ were both inverses to s , then $t = t1 = t(sr) = (ts)r = 1r = r$.
 - o A monoid or group is said to be trivial if it consists of only its identity element.
 - Therefore, if G is a nontrivial group, then it cannot have any left-zero or right-zero (or zero) element

Cartesian Product:

$$A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}.$$

- There exists $N(A) \times N(B) \times \dots \times N(K)$ elements in the n -fold cartesian product, where n is the total number of sets.
 - o Can be represented as a n -d array.
- A **subset** ($\subseteq$) includes all elements of another set, possibly being equal to it, while a **proper subset** ($\subset$) includes all elements but is always strictly smaller

Notation:

- Relation: **a subset p of $S \times T$** (the cartesian product), between two sets S, T . For $(s, t) \in p$ we write $s p t$. Thus $p = \{(s, t) \mid s p t\}$.
 - o Reflexive: the subset is formed from all $s \in S$, $s p s$. Every element is related to itself.
 - o Symmetric: the subset $s p t$ is equal to the subset $t p s$ for every $s, t \in S$. If x is related to y , then y is related to x .
 - o Antisymmetric: the subset $s p t$ is equal to the subset $t p s$ for only $(s = t) \in S$.
 - o Transitive: if $s p t$ and $t p u$ implies $s p u$ for every $s, t, u \in S$. Relations can be chained.
 - o equivalence relation on S if p is reflexive, symmetric, and transitive.
- Partition: S/p is a subset - partition, s.t. the equivalence relation p is satisfied to create a disjoint subset on S .

Surjectivity/Injectivity (bijectivity):

A mapping function is surjective if for all outputs (the codomain/range), there exists an input(s), s.t. that the entire codomain vector space is mapped to.

- If $f(S) = T$

A mapping function is injective if for all outputs, there exists exactly one input to the function such that the output is obtained.

A function that is both surjective and injective is called bijective. It is invertible.

Idempotence:

Idempotence means that the effect of an operation applied to a **set** with itself yields the original set. This property comes from the Latin *idem* (same) and *potens* (powerful), meaning "same power" or "same force".

An idempotent element is the identity element.

For any f in a finite semigroup, there exist $m < n$ such that $f^m = f^n$ hence f^{n-m} is idempotent on the image of f^m . Since in semigroup, multiplication and associativity apply.

Morphism:

Homomorphism: Of a mapping $\psi : S \rightarrow T$ such that $\psi(s_1s_2) = \psi(s_1)\psi(s_2)$ for all $s_1, s_2 \in S$ where S and T are two semigroups.

- Antihomomorphism: $\psi(s_1s_2) = \psi(s_2)\psi(s_1)$ for all $s_1, s_2 \in S$ of two semigroups S and T .

Isomorphism: If ψ is bijective, then we say that S is isomorphic to T (or S and T are isomorphic) and ψ is called an isomorphism.

- Antiisomorphism: A bijective antihomomorphism

Monoid Homomorphism: A homomorphism between monoids that does take the identity to the identity.

Transformation Semigroups, Division, Wreath Products:

- Let A be a nonvoid set. A mapping $\phi: A \rightarrow A$ is called a transformation of A .
- For every pair $s_i : A \rightarrow A$ of transformations we define the (right) multiplication s_1s_2 of s_1 by s_2 as the transformation $s : A \rightarrow A$ having $s(a) = s_2(s_1(a))$.
- Let the set S be the set of all transformations of A . This set forms a semigroup under this (right) multiplication of mappings since function composition is associative.

Transformation Semigroup: The semigroup formed by the transformation of a set A by a set of transformations H .

- $TH(A) = (A, H)$
- Note that if (A, H) is a transformation semigroup, then for all $s, s' \in H$, if for $a \in A$, $a \cdot s = a \cdot s'$ holds, then $s = s'$, since s and s' give the same transformation.
 - o Injectivity.

Elementary collapsing: A transformation s with $s(a) = s(b) = a$, but with $s(u) = u$ for all $u \in A \setminus \{a, b\}$, $a, b \in A$ with $a \neq b$.

Cyclic Permutation: A transformation ϕ such that there is a positive integer n with $a' = \phi^n(a)$, for some $a, a' \in A$.

- A cyclic permutation exists as a translation mapping on a set A with an appropriate choice of indexing for $A = \{a_1, \dots, a_n\}$ (s.t. there is a positive integer n with $a' = \phi^n(a)$).
- Cyclic permutations are defined by their length.
- A partitioned set $A \setminus B$, with a cyclic permutation $\phi|_B = \phi : A \rightarrow A$, of length $|B|$ on $B \subseteq A$, has the property $\phi(a) = a$ for every $a \in A \setminus B$ (A not in B).
- Length 2 $\Rightarrow$ transposition

$$\tau_{a,b}(u) = \begin{cases} u & \text{if } u \in A \setminus \{a, b\}, \\ a & \text{if } u = b, \\ b & \text{if } u = a \end{cases}$$

- Given a permutation $p : \{1, \dots, n\} \rightarrow \{1, \dots, n\}$, the pair $p(i), p(j)$ is called an inversion if $(p(i) - p(j))(i - j) < 0$.
 - A permutation is called **even** if the number of its inversions is even on the set A .
- $TP(A) = (A, P)$, If $A = \{1, \dots, n\}$ for some positive integer n , then P is the symmetric group of degree n .
- Alternating Group of Degree n : The **set of all even permutations** of $\{1, \dots, n\}$ forms a group under the usual (right) multiplication of mappings.
- Left vs Right transformations apply the transformation in opposite orders.

$[p^I]_{II} = p^{-1}$ and $[p^{II}]_{II} = p^{-1}$ **Corollary 1.7 (position-contents duality lemma).**

(1) Let $p_1, \dots, p_k$ be permutations on $\{1, 2, \dots, n\}$. Then

$$t_1^I \circ \dots \circ t_k^I = (t_k \circ \dots \circ t_1)^I, \quad [p_1^I \circ \dots \circ p_k^I]_{II} = p_1^{-1} \circ \dots \circ p_k^{-1} \quad \text{and} \quad [p_1^{II} \circ \dots \circ p_k^{II}]_{II} = p_k^{-1} \circ \dots \circ p_1^{-1}.$$

(2) Let $t_1, \dots, t_k$ be transformations on $\{1, 2, \dots, n\}$. Then

$$t_1^{II} \circ \dots \circ t_k^{II} = (t_1 \circ \dots \circ t_k)^{II}, \quad [t_1^I \circ \dots \circ t_k^I]_{II} = [t_k^{II} \circ \dots \circ t_1^{II}]_{II}.$$

- For a permutation (cyclic transformation) p .
- A cyclic permutation S of a set A , (A, S) is a subset of the set A s.t. the cyclic permutation is a subset or sub semigroup.
 - **Transformation Groups sets are not always subsets of the original transformed set.**

Lemma 1.8. Let A and B be two finite sets with $A \cap B \neq \emptyset$, $A \setminus B \neq \emptyset$, and let $T_G(A)$ and $T_G(A \cup B)$ denote the full permutation groups of A and $A \cup B$, respectively. Then $\langle T_G(A) \cup \{c_B\} \rangle = T_G(A \cup B)$ holds for every cyclic permutation c_B of length $|B|$ on B .

- Proved using induction.

Wreath Product: Given two transformation semigroups (X, S) and (Y, T) :

$$(Y, T) \wr (X, S) = (Y \times X, W)$$

where W is the set of all transformations w of $Y \times X$ satisfying for all $(y, x) \in Y \times X$:

$$(y, x) \cdot w = (y \cdot f(x), x \cdot s)$$

for some fixed $f : X \rightarrow T$ and $s \in S$, both depending only on w . For each element w of W , f and s are uniquely determined. Therefore, W is the set of all (f, s) with $f : X \rightarrow T$ and $s \in S$.

$$(y, x) \cdot ((f, s)(g, s')) = (y \cdot f(x), x \cdot s)(g, s') = (y \cdot f(x)g(x \cdot s), x \cdot ss').$$

The composition above implies that W has closure under composition $\Rightarrow W$ is indeed a semigroup. $\Rightarrow$ The wreath product $(Y \times X, W)$ is a transformation semigroup.

- If S and T are monoids, then so is W
- If (X, S) and (Y, T) are permutation groups, then so is their wreath product
- the wreath product is an associative operation on the class of transformation semigroups and also on the class of permutation groups.

Graphs:

- Directed graph has direction.

Theorem 2.1 (Kuratowski planar graph theorem). *A graph G is planar if and only if it contains no subdivision of K_5 , the complete graph with five vertices, nor of the complete bipartite graph $K_{3,3}$.* $\square$

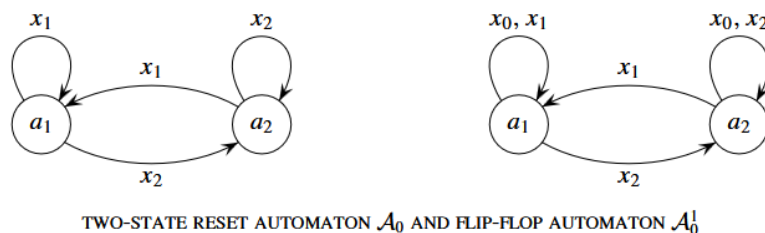
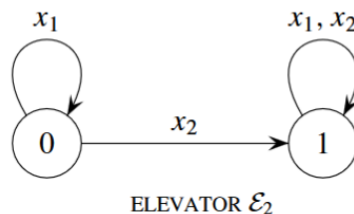
Automata and Automata Networks:

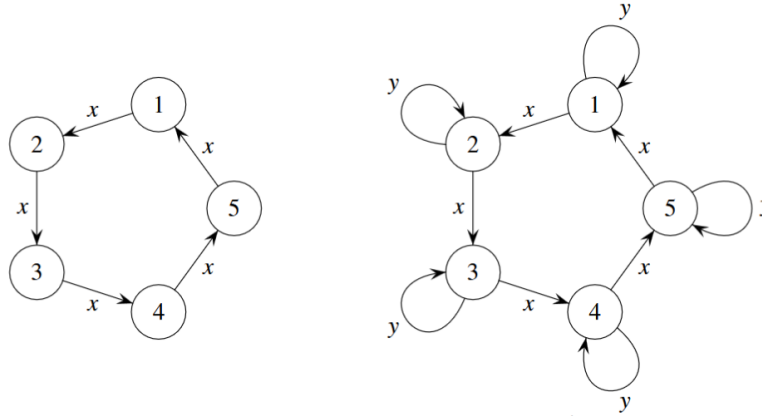
Automaton: $A = (A, X, \delta)$ is a finite automaton without outputs. Here A is the (finite nonempty) state set, X is the input alphabet, and $\delta : A \times X \rightarrow A$ is the transition function.

- **Singleton (trivial automaton):** there exists only one state $a \in A$
- Autonomous automaton: $a \in A$, $x, y \in X$ and the equality $\delta(a, x) = \delta(a, y)$ holds. The input alphabet is time (not considered) and hence all $x, y \in X$ has same effect by δ .
- Permutation Automaton: An invertible automaton with a transformation $\delta_x : a \rightarrow \delta(a, x)$ ($a \in A$, $x \in X$) such that the mapping is a one-to-one mapping of A onto itself.

- Reset Automaton (Constant map automaton): An automaton which for all $\{\delta(a, x) \mid a \in A, x \in X\}$, the output/set is the same state a_k (a singleton).
 - o Identity is a reset automaton if $\delta = 1$.
 - o Constant map c (a transition, subset of characteristic semigroup):
 - $c^2 = c$ (idempotent)
 - $\ker(c) = Q \times Q$ (everything collapses)
 - **maximum possible information loss**
- Permutation-Reset Automaton: An automaton whose transformation δx is either a constant map or a permutation.
- Identity-Reset Automaton: An automaton whose transformation δx is either a constant map or the identity permutation.
- Generators in an Automaton A: the subset B of A s.t. for every $a \in A$ there are $b \in B$ and $p \in X^*$ fulfilling $\delta(b, p) = a$.
- Connected Automaton: it has a state a such that for every state b there exists an input word p having $\delta(a, p) = b$. Then we also say that A is connected for the state a .
 - o strongly connected if for every pair $a, b \in A$ of states there is a word $p \in X^*$ with $\delta(a, p) = b$.
- Dead state: A state $a \in A$ such that for every pair $b \in A, x \in X$ and positive integer $m \geq n$, $\delta(b, x^m) = a$.
- Nilpotent: For a dead state $a \in A$, every pair $b \in A, p \in X^*, |p| \geq n$, $\delta(b, p) = a$.
- Commutative: for any state $a \in A$ and input words $p, q \in X^*$, $\delta(a, pq) = \delta(a, qp)$

Automatons as represented by mealy state machines (2 state automaton):





COUNTER C_5 AND COUNTER WITH IDENTITY C_5^1

Characteristic Semigroup: of an automaton A , $S(A) = \{\delta p \mid p \in X^+\} \Rightarrow$ the transition function s.t. for every δp , $\delta q \in S(A)$, so that $\delta p \delta q = \delta pq$. Associativity holds.

Characteristic Monoid: of an automaton A (characteristic semigroup), where the identity element transition function $\delta \lambda$, where $\lambda \in X^*$ exists.

Permutation Automaton: of an automaton A (characteristic monoid), that is invertible. An invertible transition function means that the automaton is a group and is thus a permutation automaton and cyclic.

- trivial if its characteristic monoid is a singleton

Subautomaton: $A' = (A', X', \delta')$ of the automaton $A = (A, X, \delta)$ s.t. $A' \subseteq A$, $X' \subseteq X$ and δ' is the restriction of δ to $A' \times X'$.

- Set of all words: $X^* \subseteq X$
- Non-empty words: $X^+ \subseteq X$

Homomorphism: If an automaton $A = (A, X, \delta)$ has surjective mappings $\psi_1 : A \rightarrow A'$, $\psi_2 : X \rightarrow X'$ onto $A' = (A', X', \delta')$ so that for a state $a' \in A'$ (state space) $a' = \psi_1(\delta(a, x)) = \delta'(\psi_1(a), \psi_2(x))$.

- If ψ_1 and ψ_2 are bijective functions, then A is said to be isomorphic to A' .

Compatible Transformation on digraph: A transformation $F : X^n \rightarrow X^n$ with a digraph $D = (V, E)$ (of order n) if F has the form $F(x_1, \dots, x_n) = (f_1(x_1, \dots, x_n), \dots, f_n(x_1, \dots, x_n))$ $((x_1, \dots, x_n) \in X^n)$, and $f_i : X^n \rightarrow X$, $i = 1, \dots, n$ may depend only on x_i and those x_j for which $(v_j, v_i) \in E$ (including the case $v_i = v_j$). If F is compatible with D , then sometimes we also say that F is D -compatible.

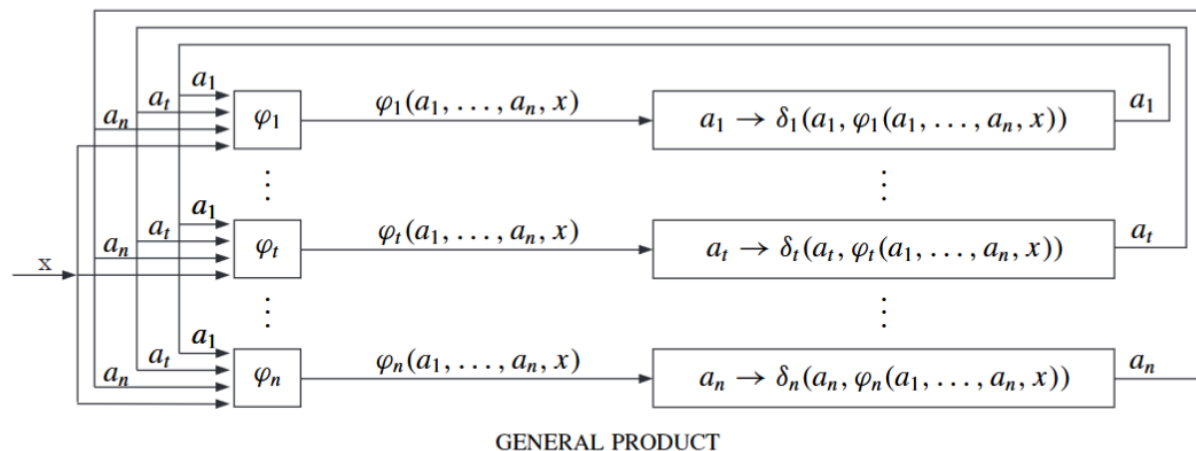
Finite Automata Network: An automaton $A = (A, X, \delta)$, let $A = A_1 \times \dots \times A_n$ for some $|A_i| \geq 1$ and $n \geq 1$.

- D-network: For a Finite Automata Network $A = (A, X, \delta)$, let $A = A_1 \times \dots \times A_n$ s.t. δ_x : $A \rightarrow A$ ($x \in X$) is compatible with D.

Size-n Automata Network: Consisting of n component automata $A_i = (A_i, A_1 \times \dots \times A_n \times X, \delta_i)$, $i \in \{1, \dots, n\}$, such that the size- n automata network is an automata consisting of n automata networks. Where the δ_i are defined by $\delta(a, x) = (\delta_1(a_1, (a, x)), \dots, \delta_n(a_n, (a, x)))$ for $a = (a_1, \dots, a_n) \in A$, $a_i \in A_i$, $x \in X$. $\Rightarrow$ Compatibility.

General Product (Gluskov-type): The Automaton $\mathbf{A} = (\mathbf{A}_1 \times \dots \times \mathbf{A}_n) (X, (\phi_1, \dots, \phi_n))$, which is the product between the system automata network of size n and feedback functions ϕ_i ($i \in \{1, \dots, n\}$). The product has properties such that the state set is $\mathbf{A} = A_1 \times \dots \times A_n$, input set X , and transition function δ given by $\delta((a_1, \dots, a_n), x) = (\delta_1(a_1, \phi_1(a_1, \dots, a_n, x)), \dots, \delta_n(a_n, \phi_n(a_1, \dots, a_n, x)))$ for all $(a_1, \dots, a_n) \in A$, and $x \in X \Rightarrow$ compatible.

- Let $A_i = (A_i, X_i, \delta_i)$ be automata where $i \in \{1, \dots, n\}$
- Feedback function $\phi_i : A_1 \times \dots \times A_n \times X \rightarrow X_i$ for every $i \in \{1, \dots, n\}$.



K-R Preliminaries:

Collapsing (of an automaton): If the transition semigroup of an automaton $A = (A, X, \delta)$ has $\delta(a_1) = \delta(a_2) = \dots = \delta(a_n) = r$, $a \in A$ (state set), then the transition δ is not injective. Thus, the preimage of r is not distinguishable. Elementary collapsing generates the elementary quotient a_i s.t. the preimage of r is a_i .

- Non-injective transitions force collapsing
- Injective transitions survive as reversible control

Division: S divides T if for some $T' \subseteq T$, the semigroup S is a homomorphic (surjective by definition) image of a sub semigroup of T s.t. $\phi: T' \rightarrow S$

S divides an Automaton A if S divides the characteristic semigroup $S(A)$ of A (transition semigroup).

$$S \overset{\varphi}{\leftarrow} T \leq S(A)$$

- Length condition: $T' \subseteq T$ is a transformation coming from a word of length n . "So all elements of S are represented **at the same "time depth"**."
 - o $S \parallel (n)A$
 - o For a word: $w = x_1 x_2 \cdots x_n = x_n \circ \cdots \circ x_1$ with $x_i \in X$, define: $\delta_{word} = \delta_{x_n} \circ \cdots \circ \delta_{x_1}$.
- When we say a semigroup **S divides** another semigroup T , we mean:
- $S \cong$ homomorphic image of a subsemigroup of T .

Embeds: If $S < T$ (divides - division) s.t. $\phi: T \rightarrow S$ for $T' \subseteq T$ and if ϕ is a one-to-one mapping, then

we also say that S embeds in $S(A)$ in equal lengths n with respect to B .

- Note: Notice that for every monoid M and automaton A , we have $M \parallel A$ if and only if $M \parallel S(A^\lambda)$.

Kernel (of a function): Let $f: Q \rightarrow Q$ be any function (not necessarily homomorphic), the kernel of f is an equivalence relation: $\ker(f) = \{(p, q) \in Q \times Q \mid f(p) = f(q)\}$. The equivalence relation defines all not one-to-one preimages with the same image (the idempotent set is a subset of this set).

- The kernel is directly correlated with the idempotent e .
- The equivalence relationship applied, $Q/\ker(e)$, gives a sets (/equivalence classes) corresponding to **one image point of f/e** .
- $Q/\ker(e) = \{e^{-1}(r) \mid r \in \text{Im}(e)\}$

Example:

Let

$$Q = \{A, B, C, D\}$$

and

$$e(A) = C, e(B) = C, e(C) = C, e(D) = D$$

Then:

- $\text{Im}(e) = \{C, D\}$

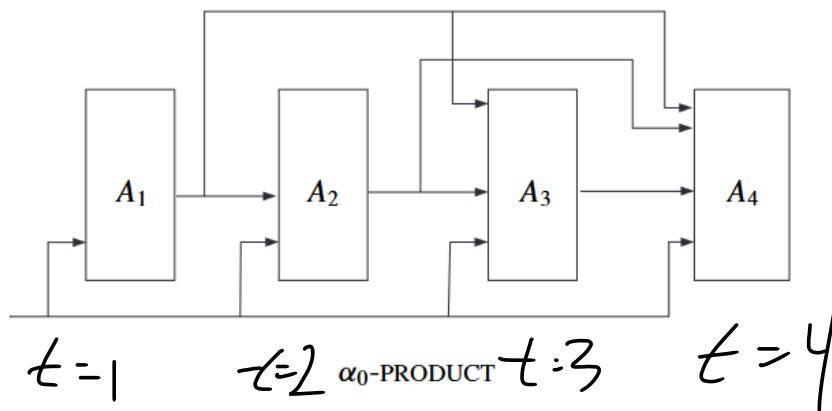
And:

$$Q / \ker(e) = \{\{A, B, C\}, \{D\}\}$$

An elementary quotient is the quotient of an automaton (or semigroup action) by the kernel of a single idempotent transformation.

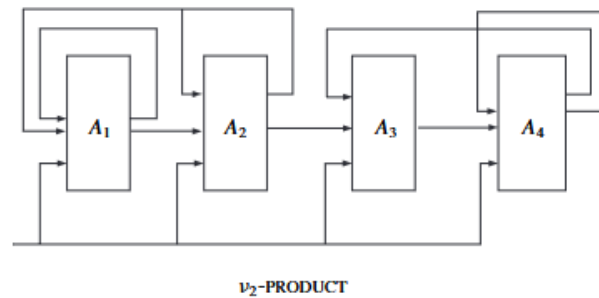
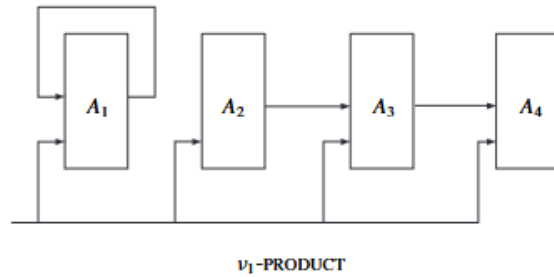
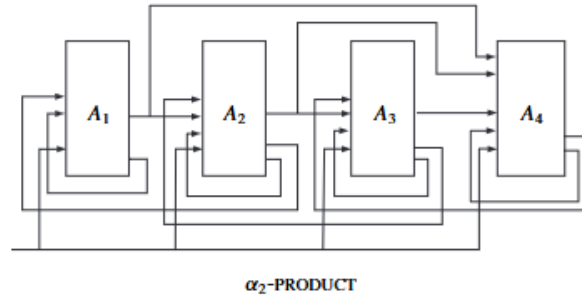
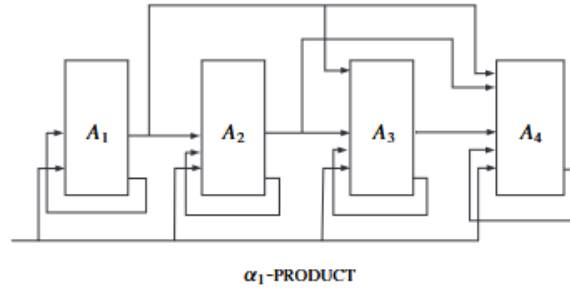
Cascade product (α_0 -product): of an automaton A , where for every $i \in \{1, \dots, n\}$, ϕ_i really may not depend on its j th state variable if $j \geq i$.

$$\phi_1 : X \rightarrow X_1, \phi_2 : A_1 \times X \rightarrow X_2, \dots, \phi_n : A_1 \times \dots \times A_{n-1} \times X \rightarrow X_n$$



α_i -product: ($i = 0, 1, \dots$); if each ϕ_t ($t = 1, \dots, n$) is really independent of its j th state component ($j = 1, \dots, n$) whenever $j \geq t + i$.

- If i is a positive integer for which every ϕ_t ($t = 1, \dots, n$) really depends on not more than i state variables, then A is a ν_i -product. In addition, an $\alpha_i - \nu_j$ -product ($i = 0, 1, \dots, j = 1, 2, \dots$) is an α_i -product that is also a ν_j -product



Flip-Flop Monoid Revisited: A monoid is a set where associativity holds, and the identity element exists. FF is the smallest monoid s.t. there exists two input actions 1: the identity and a: the reset (maps to a). Due to idempotence, and only two input actions, it is noninvertible on input a. The flip flop monoid is produced by elementary collapsing of a non-injective idempotent transformation $f: Q \rightarrow Q$ with $e = f^k$ s.t. $e(s) = r, s \in S$ that collapses states. The sub monoid generated by $\{1, e\}$ is isomorphic to FF monoid, with irreversibility.

Character wise maps: maps that show an automaton's states to an input word.

Truncated run maps: maps that show an automaton's states to an input word with elementary collapses on idempotent images removed such that the run map is made invertible.

Maximal subgroup: Let S be a finite semigroup and let $e \in S$ be idempotent. The set/group $H_e = \{ s \in S \mid se = es = s \}$.

Krohn-Rhodes Theorem:

Krohn-Rhodes Theorem allows the decomposition of an automaton into permutation (reversible group behavior) and reset (irreversible flip flop monoid behavior) automata which demonstrate the character wise & truncated word maps of the automata.

- Of an automaton, $S(A)$ (transformation semigroups), there exists uninvertible transformations, idempotents $\Rightarrow$ information loss, elementary collapses.
- Wreath products combine permutation and reset automata into a cascade product...?
- Coordinate systems are layered so that reversibility is controlled by higher level coordinates on irreversible transitions.
 - "Each coordinate corresponds to a factor in the wreath product decomposition of the transition semigroup, and each factor acts on its own coordinate space."
 - "Evolutionary emergence: flip-flop layers record irreversible evolutionary decisions; group layers record structured variation that survives within those decisions."